Game-theoretic Analysis of Privacy Preservation in Location-Based Services

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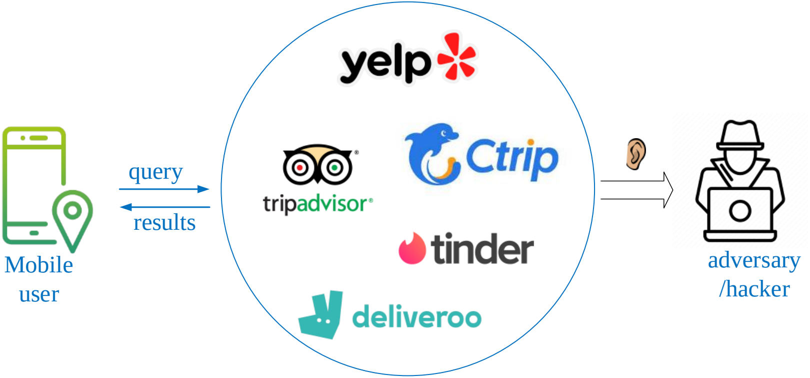
Overview

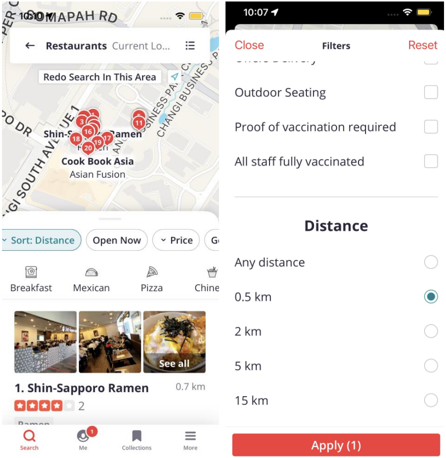
Privacy Issues in Location-Based Services (LBSs)



LBS platforms to find nearby Points-of-Interests (PoIs, e.g., restaurants, hotels)

User's Service Flexibility

In practice, many users are actually flexible in LBS service requirement. They are not only interested in the information of Point-of-Interest (PoI) exactly at their real locations but also those PoIs around them [T. Li et al., 2009].



Key Questions

Question 1

Whether a user can gain privacy with service flexibility, given he only reports one or a few (dummy) locations to the LBS platform?

Question 2

How a user should best gain location privacy from his service flexibility, by optimally reporting his dummy locations?

- Scheme 1: Multi-query Bayesian Game
- Scheme 2: Multi-user Privacy Cooperation Game

Scheme 1: Multi-query Bayesian Game [1]

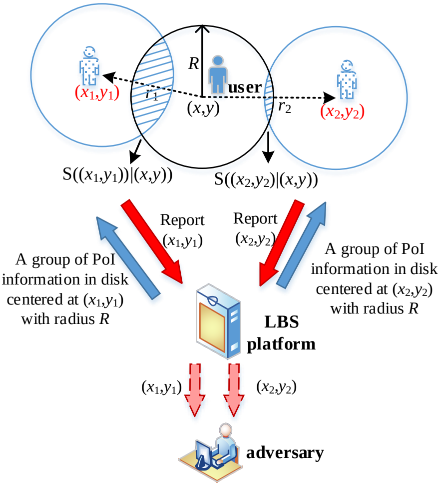
- Location privacy preservation from LBS service flexibility
- Two-stage Bayesian game theoretic approach against an intelligent adversary
- Performance gain over traditional k-anonymity mechanisms, especially under a small number k

[1] S. Hong, L. Duan and J. Huang, "Protecting Location Privacy by Multi-Query: A Dynamic Bayesian Game Theoretic Approach," IEEE Transactions on Information Forensics and Security (TIFS), vol. 17, pp. 2569-2584, 2022.

Related Work

- Study on the tradeoff between privacy and service utility
  - Privacy-preserving mechanisms for crowdsensing [Y. Zhang et al., 2020], multi-agent [C. X. Wang et al., 2020] and LBS systems [T. Feng et al., 2019]
- Game-theoretic analysis between a user and an adversary
  - Location privacy preservation by reporting location with a certain level of granularity [W. Wang et al., 2015]
  - Optimal strategy against adversary's localization attacks [R. Shokri et al., 2016], [L. Xu et al. 2018]

System Model: Dual-Query



- The user with real location (x, y) reports (x_1, y_1) and (x_2, y_2) on the LBS platform.
- The platform provides the user with a group of nearby PoI information in the circular disks centered at (x_1, y_1) and (x_2, y_2) with radius R, respectively.
- After the user's geo-data reports, the adversary has probability η to access both geo-data and $1 - \eta$ to access only one.

System Model: Adversary's Inference Attack of User's Real Location

- We consider an intelligent attack, where the adversary has full information about the user's payoff function, thus it can estimate the user's reporting strategies and optimally infer the user's real locations.
- The adversary's objective is to minimize the inference error (i.e., expected Euclidean distance from the guessed location to user's real location).

System Model: Adversary's Inference Attack of User's Real Location

- Depending on the number of geo-data that the adversary can access, it will infer the user's real location (x, y) from user's reported geo-data (x_1, y_1), or (x_2, y_2), or both.
  - If the adversary can only access one reported location, i.e., geo-data (x_i, y_i) $(i \in \{1, 2\})$:

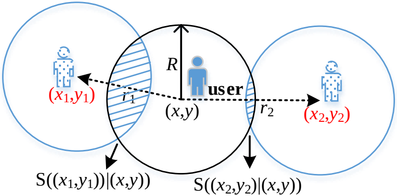
$$\begin{aligned}&\left(\hat{x}_i^{BNE}(x_i, y_i), \hat{y}_i^{BNE}(x_i, y_i)\right)\\&= \arg\min_{(x_i', y_i')} \mathbb{E}_{(X,Y)} d\left((x_i', y_i'), (X, Y)\right),\end{aligned} \tag{1}$$

    where (X, Y) are 2D random variables denoting the possible locations for the real location (x, y) given the user's reported geo-data (x_i, y_i).
  - If the adversary can access both reported locations:

$$\begin{aligned}&\left(\hat{x}^{BNE}((x_1, y_1), (x_2, y_2)), \hat{y}^{BNE}((x_1, y_1), (x_2, y_2))\right)\\&= \arg\min_{(x', y')} \mathbb{E}_{(X,Y)} d\left((x', y'), (X, Y)\right),\end{aligned} \tag{2}$$

    where (X, Y) are 2D random variables denoting the possible locations for (x, y) given two geo-data (x_1, y_1) and (x_2, y_2).

System Model: User's Strategic Location Reporting



We consider the user strategically reports two geo-data (x_i, y_i) ($i \in \{1, 2\}$) on the platform i.

- $$r_i = d\left((x_i, y_i), (x, y)\right) = \sqrt{(x_i - x)^2 + (y_i - y)^2}$$

  denotes the reporting distance, i.e., the Euclidean distance between the reported location and the real one.

- $$S((x_1, y_1), (x_2, y_2)) = S((x_1, y_1)|(x, y)) \cup S((x_2, y_2)|(x, y))$$

  denotes the total service coverage area by querying the LBS platforms.

System Model: User's Payoff

User's Payoff U = Privacy Gain $\times$ Service Gain

- **User's Service Gain**

We consider that the locations of PoIs follow a 2D Poisson point process (PPP) with a **spatial density** λ. The random variable $N(\lambda, S)$ to denote **the total number of PoIs** in the service coverage area S. The probability mass function (PMF) of having N PoIs is

$$\Pr(N = n) = \frac{(\lambda S)^n}{n!} \exp\left(-\lambda S\right), \quad n = 0, 1, \cdots,$$

where the mean of $N(\lambda, S)$ is $\bar{N} = \lambda S$, which increases with the service coverage area S and the PoI density λ.

Definition (user's service gain function)

The user's service gain function $f(\bar{N})$ is an increasing and concave function in the mean number $\bar{N} = \lambda S$ of the searched PoIs. The function satisfies $f(0) = 0$, $f(\bar{N}) \geq 0$, $f'(\bar{N}) > 0$ and $f''(\bar{N}) < 0$ for $\bar{N} \geq 0$.

System Model: User's Payoff

User's Payoff U = Privacy Gain $\times$ Service Gain

- **User's Privacy Gain**

The user's privacy gain depends on whether the adversary can access one or both geo-data, as well as the adversary's attack strategy of inferring the user's location. We define the **privacy gain** as the minimum Euclidean distance from the guessed location to the user's real location:

$$\text{Privacy gain} = \begin{cases} \min_{i=1,2} d\left((\hat{x}_i^{BNE}, \hat{y}_i^{BNE}), (x, y)\right), \\ \text{if adversary accesses one geo-data,} \\ d\left((\hat{x}^{BNE}, \hat{y}^{BNE}), (x, y)\right), \\ \text{if adversary accesses both geo-data.} \end{cases}$$

System Model: User's Payoff

User's Payoff U = Privacy Gain $\times$ Service Gain

- In reality, the user does not know whether the adversary accesses one or both geo-data.
- Assume the user holds the belief that the adversary accesses both geo-data with an **access probability** η and one geo-data with a probability $1-\eta$.
- **User's expected payoff:**

$$
\begin{aligned}
&U((x_1,y_1),(x_2,y_2)|(x,y)) \\
=&\left((1-\eta)\min_{i=1,2} d\left((\hat{x}_i^{BNE},\hat{y}_i^{BNE}),(x,y)\right)+\eta d\left((\hat{x}^{BNE},\hat{y}^{BNE}),(x,y)\right)\right) \\
&\times f\left(\lambda S((x_1,y_1),(x_2,y_2))\right).
\end{aligned}
$$

System Model: Two-Stage Game

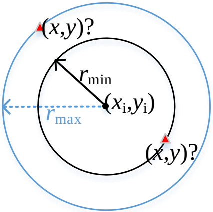
- In Stage I, the user makes a deterministic strategy on how to report two locations (x_1, y_1) and (x_2, y_2) on the platforms, given his real location (x, y).
- In Stage II, the adversary infers the user's location as $(\hat{x}_i, \hat{y}_i)$ in (1) or $(\hat{x}, \hat{y})$ in (2), depending on whether it accesses one or both geo-data.

$\star$ Analyze the two-stage game by Backward Induction.

Stage II: Adversary's Best Inference Attack

Proposition

When the adversary can only access the user's one geo-data report (x_i, y_i), its best inference attack at the Bayesian Nash Equilibrium (BNE) is the same as the geo-data report, i.e., $(\hat{x}_i^{BNE}, \hat{y}_i^{BNE}) = (x_i, y_i)$.

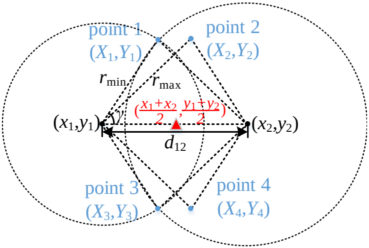


- When the adversary only has access to (x_i, y_i), all possible location candidates for (x, y) appear on the two circles, and are either distance $r_{\min}$ or $r_{\max}$ away from the circles' center (x_i, y_i).

Stage II: Adversary's Best Inference Attack

Proposition

When the adversary can access the two geo-data (x_1, y_1) and (x_2, y_2), its best inference attack at the BNE is the middle point of the user's two reported dummy locations, i.e., $(\hat{x}^{BNE}, \hat{y}^{BNE}) = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.



- When the adversary has access to both (x_1, y_1) and (x_2, y_2), there are four candidates for (x, y) to appear at the intersection points of the two circles centered at the geo-data (x_1, y_1) and (x_2, y_2) with radius $r_{\min}$ or $r_{\max}$.

Stage I: User's Best Strategy in One-dimension (1D)

Theorem (User's Equilibrium Reporting Strategy in 1D)

In the 1D linear line, there exists a threshold $\eta_{th} \in (0,1)$ such that the user's equilibrium reporting strategy of (x_1^{BNE}, x_2^{BNE}) depends on η as follows.

- *If the adversary has a small probability to access both geo-data, i.e., $0 \leq \eta \leq \eta_{th}$, the user will strategically report geo-data $(x_1^{BNE} = x \mp r^{**}, x_2^{BNE} = x \pm r^{**})$ on two opposite sides of x, where r^{**} is a constant distance.*
- *If the adversary has a large probability to access both geo-data, i.e., $\eta_{th} < \eta \leq 1$, the user will report $(x_1^{BNE} = x \pm r_1^*(\eta), x_2^{BNE} = x \pm L)$[a] on the same side of x, where $r_1^*(\eta)$ is a decreasing function in η.*

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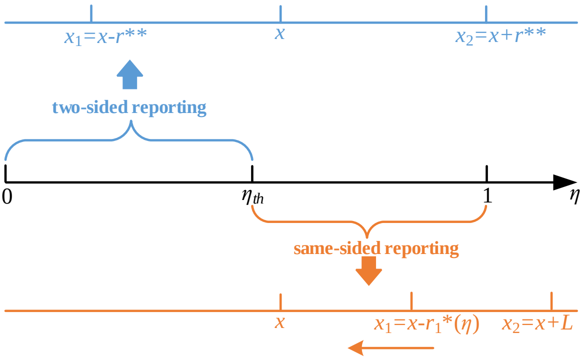
[a] As the user cannot report a dummy location arbitrarily far away from x to confuse the adversary, we limit the reporting distance by L to avoid the trivial case.

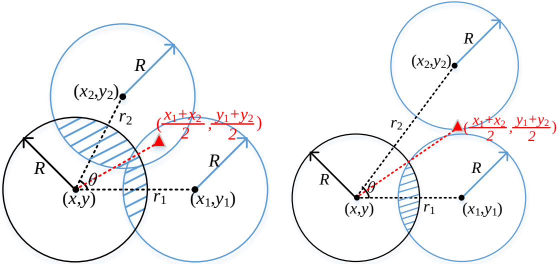
User's Threshold-based Reporting Strategy in 1D



- As the adversary's probability η to hack into both platforms (in user's belief) increases from 0 to 1, the user changes from two-sided symmetric location reporting to the same-sided reporting once η exceeds threshold η_{th}.

Stage I: User's Best Strategy in Two-dimension (2D)

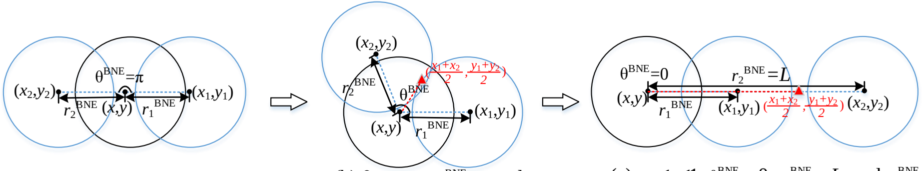
Based on trigonometry, we can uniquely specify the user's reporting strategy through three decision variables (r_1, r_2, θ).



(a) Utilize both (x_1, y_1) and (x_2, y_2) for positive service coverages.

(b) Utilize only (x_1, y_1) for the service gain and (x_2, y_2) for the privacy gain.

User's Threshold-based Reporting Strategy in 2D



(a) $\eta=0$: $\theta^{BNE}=\pi$ and $r_1^{BNE}=r_2^{BNE}$.

(b) $0<\eta<\eta_{th}$: $\theta^{BNE}<\pi$ and decreases with η.

(c) $\eta_{th}\leq\eta\leq 1$: $\theta^{BNE}=0$, $r_2^{BNE}=L$, and r_1^{BNE} decreases with η.

The user's reporting strategy in 2D also follows a threshold structure:

- The user chooses a different-sided reporting strategy ($0<\theta^{BNE}<\pi$) with a service advantage when $\eta<\eta_{th}$
- The user chooses the same-sided reporting strategy ($\theta^{BNE}=0$) with a privacy advantage when $\eta>\eta_{th}$

Performance Improvement

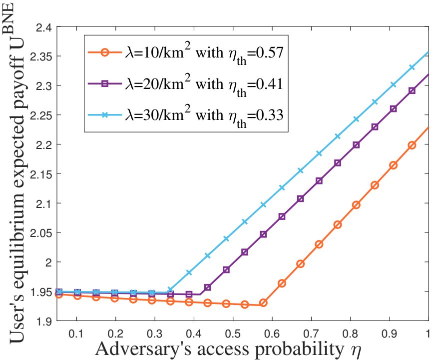


Figure: User's equilibrium expected payoff U^{BNE} versus η. The parameter setting is $R = 1$ km, $\lambda \in \{10, 20, 30\}/\text{km}^2$, $L = 3$ km.

- The user's equilibrium expected payoff U^{BNE} first decreases with $\eta < \eta_{th}$ under the different-sided reporting strategy, and then surprisingly increases with $\eta \geq \eta_{th}$ under the same-sided reporting strategy.

Performance Improvement: Multi-Query

k	1	2	3	4	5	6	7	8
k**-anonymity**	0	0.34	0.69	1.02	1.36	1.68	2.01	2.36
Multi-query	1.54	2.72	2.99					

Table: User's payoff under k-anonymity and our multi-query approach (when the adversary accesses all geo-data). The parameter setting is $R = 1$ km, $\lambda = 10/\text{km}^2$, $L = 4$ km.

- The user's payoff improvement between $k = 2$ and $k = 3$ is merely 0.27, showing that the payoff improvement becomes less obvious in k with a diminishing return with more queries.

Scheme 2: Multi-user Privacy Cooperation Game [2]

- Preserve user's privacy under a multi-user cooperation scheme
- Take a no-hiding strategy, i.e., always querying the LBS platform even when a user gets help from his peers
- Formulate multi-user privacy cooperation as a max-min adversarial game problem

[2] S. Hong and L. Duan, "Multi-user Privacy Cooperation Game by Leveraging Users' Service Flexibility," Proc. IEEE International Symposium on Information Theory (ISIT), 2022.

Related Work

- **Caching-based approaches**
  - Single-user cache: [Zhu et al. 2013], [Niu et al. 2015], [Zhang et al. 2019]
    - Locally cache a user's historical query information and skip future similar LBS queries leaking his privacy:

  However, an individual user's caching is far from enough to cover many PoIs to visit later, and it is promising for multiple users to share their queried PoIs with each other in the common cache.

Related Work

- **Caching-based approaches**
  - Multi-user cache with single-query: [Shokri et al. 2014]
    - Only if the user finds the shared PoI information in the cache not useful, he queries the LBS using his real location, which might also be compromised by the adversary

  To improve the case that a user does not hear any useful PoIs from others in the cache:
  - Multi-user cache with multi-query: [Peng et al. 2017], [Hu et al. 2018], [Jung et al. 2017], [Cui et al. 2020]
    - Allow a user to make multiple queries and bear extra complexity/overhead, via k-anonymity or l-diversity
    - Only work well for a large number k and are also vulnerable to attacks with background knowledge [Rajendran 2017]

Related Work

- Study on the tradeoff between privacy and service utility
  - Privacy-preserving mechanisms for crowdsensing [Y. Zhang et al., 2020], multi-agent [C. X. Wang et al., 2020] and LBS systems [T. Feng et al., 2019]
- Game-theoretic analysis between a user and an adversary
  - Location privacy preservation by reporting location with a certain level of granularity [W. Wang et al., 2015]
  - Optimal strategy against adversary's localization attacks [L. Xu et al. 2018]. Some works further models this flexibility as a service requirement constraint [R. Shokri et al., 2016].

Multi-user location privacy game

Key Question 1

If a user already finds the shared PoI information useful in the cache, is it beneficial for him to hide from or further query the LBS platform?

Key Question 2

If a user finds the shared PoI information not useful in the cache, how to strategically add more obfuscation to his query?

Key Question 2

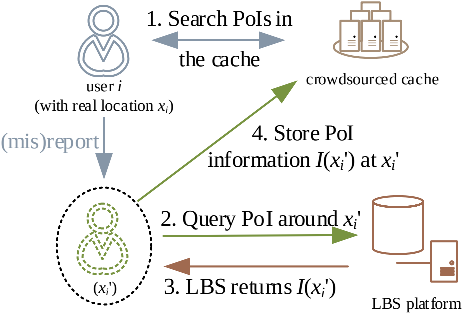
What is the adversary's best inference attack to the multi-user strategic LBS querying?

- Preserve user's privacy under a multi-user cooperation scheme
- Take a no-hiding strategy, i.e., always querying the LBS platform even when a user gets help from his peers
- Formulate multi-user privacy cooperation as a max-min adversarial game problem

System Model

- We consider N LBS users in a set $\mathcal{N} = \{1, 2, \cdots, N\}$ sequentially arriving with their real locations $x_1, x_2, \cdots, x_N$ in a continuous bounded location set $\mathcal{M}$.
- Each user $i \in \mathcal{U}$ wants to query the LBS server for nearby PoI information, by reporting a location $x_i' \in \mathcal{M}$.
- The LBS server will return PoI information $I(x_i')$ exactly at x_i' to the user, and the user only finds the PoI within a distance Q_i useful.

System Model under the Cooperative Scheme



- Step 1: user i searches the cooperative cache created by previous users, for possible PoI information around x_i.
- Step 2: user i may continue to query the LBS server with a location x_i', which may or may not be the same as his real location x_i.
- Step 3: the LBS se
- Step 4: user i stores the reportrver returns PoI information $I(x_i')$ to the user. ed location x_i' and the PoI information $I(x_i')$ in the cache to benefit latter users.

User's Reporting Strategy

Suppose user i is the i-th among N users to arrive, then he may or may not benefit from the cached information $I(x_j')(j = 1, \cdots, i-1)$, leading to different reporting strategies.

- User i benefits from the cache if he finds any user j's shared $I(x_j')$ within Q_i distance, i.e., $\exists d(x_i, x_j') \leq Q_i$. Within the region

$$\mathcal{X}_i^{in}(\boldsymbol{x}_{i-1}') = \{x \in \mathcal{M} | \min_{j=1,\cdots,i-1} d(x, x_j') \leq Q_i\},$$

  user i's service constraint is satisfied.
- Otherwise, if x_i is within the region

$$\mathcal{X}_i^{out}(\boldsymbol{x}_{i-1}') = \{x \in \mathcal{M} | \min_{j=1,\cdots,i-1} d(x, x_j') > Q_i\}$$

  user i's reporting strategy should be designed to meet user i's service quality constraint: $d(x_i, x_i') \leq Q_i$, $\forall x_i \in \mathcal{X}_i^{out}$.
- Particularly, the first-arriving user 1 finds the cache empty with $\mathcal{X}_1^{IN} = \emptyset$.

User's Reporting Strategy

Definition (User i's general reporting strategy)

Depending on whether the cached PoIs are useful, we can generally define user i's reporting strategy to LBS as

$$f_i(x_i'|x_i, \boldsymbol{x}_{i-1}') = \begin{cases} f_i^{in}(x_i'|x_i, \boldsymbol{x}_{i-1}'), & \text{if } x_i \in \mathcal{X}_i^{in}(\boldsymbol{x}_{i-1}'), \\ f_i^{out}(x_i'|x_i, \boldsymbol{x}_{i-1}'), & \text{if } x_i \in \mathcal{X}_i^{out}(\boldsymbol{x}_{i-1}'). \end{cases}$$

after observing $I(x_j')$ for any $j = 1, \cdots, i-1$ in the cache.

User's Expected Privacy Gain

Assuming that the adversary will infer user i's real location as $\hat{x}_i(x_i'|\boldsymbol{x}'_{i-1})$ after observing user i's reporting x_i' as well as all former users' queries $\boldsymbol{x}'_{i-1}$, we give user i's expected privacy gain as:

$$\begin{aligned}
\pi_i = & \int_{x_i'} \int_{x_i \in \mathcal{X}_i^{in}(\boldsymbol{x}'_{i-1})} \psi_i(x_i) f_i^{in}(x_i'|x_i, \boldsymbol{x}'_{i-1}) \\
& \qquad d(x_i, \hat{x}_i(x_i'|\boldsymbol{x}'_{i-1})) dx_i dx_i' \\
& + \int_{x_i'} \int_{x_i \in \mathcal{X}_i^{out}(\boldsymbol{x}'_{i-1})} \psi_i(x_i) f_i^{out}(x_i'|x_i, \boldsymbol{x}'_{i-1}) \\
& \qquad d(x_i, \hat{x}_i(x_i'|\boldsymbol{x}'_{i-1})) dx_i dx_i' \\
= & \int_{x_i'} \int_{x_i} \Pr(x_i, x_i'|\boldsymbol{x}'_{i-1}) \\
& \qquad d(x_i, \hat{x}_i(x_i'|\boldsymbol{x}'_{i-1})) dx_i dx_i',
\end{aligned}$$

where $\psi_i(x_i)$ is user i's location distribution.

Adversary's Optimal Inference Attack

- To provide robust privacy protection for users, we consider the challenging case of an adversary with strong information.
  - The adversary has full information about all users' privacy objective functions, location distribution $\psi_i(x_i)$ $(i = 1, \cdots, N)$, and can also access the cache and LBS server.
  - Thus it can estimate users' reporting strategies $\boldsymbol{f}_i(x_i'|x_i)$ on their behalves, and optimally infer users' real locations.

Adversary's Optimal Inference Attack

- After observing all users' queries, the posterior distribution of user i's real location to the adversary is a function of $(\boldsymbol{x}'_{i-1}, x'_i)$:

$$\Pr(x_i|x'_i, \boldsymbol{x}'_{i-1}) = \frac{\Pr(x_i, x'_i|\boldsymbol{x}'_{i-1})}{\Pr(x'_i|\boldsymbol{x}'_{i-1})}$$
$$= \frac{\psi_i(x_i) f_i(x'_i|x_i, \boldsymbol{x}'_{i-1})}{\int_{x_i \in \mathcal{M}} \psi_i(x_i) f_i(x'_i|x_i, \boldsymbol{x}'_{i-1}) dx_i}.$$

- To minimize the inference error to user i's real location x_i, the adversary's optimal inference problem for user i is given as:

$$\min_{\hat{x}_i} \int_{x_i} \Pr(x_i|x'_i, \boldsymbol{x}'_{i-1}) d(\hat{x}_i, x_i) dx_i.$$

Max-min Problem Formulation

- Two-stage game
  - In Stage I, user i makes a probabilistic strategy $\boldsymbol{f}_i(x_i'|x_i, \boldsymbol{x}_{i-1}') = \{f_i^{in}(x_i'|x_i, \boldsymbol{x}_{i-1}'), f_i^{out}(x_i'|x_i, \boldsymbol{x}_{i-1}')\}$ on how to report location to the LBS server to maximize his privacy gain, considering whether he finds the cached PoI information useful or not.
  - In Stage II, the adversary infers the user i's location as $\hat{x}_i$ after observing the reported location x_i' in LBS as well as former users' queries $\boldsymbol{x}_{i-1}'$, to minimize its inference error.

User i's Expected Privacy Gain

- Considering the adversary's optimal inference and averaging the adversary's inference over all possible x_i''s:

$$\begin{aligned}
\pi_i^{opt} &= \int_{x_i'} \Pr(x_i' | \boldsymbol{x}'_{i-1}) \min_{\hat{x}_i} \int_{x_i} \Pr(x_i | x_i', \boldsymbol{x}'_{i-1}) d(\hat{x}_i, x_i) dx_i dx_i' \\
&= \int_{x_i'} \left(\min_{\hat{x}_i} \int_{x_i} \Pr(x_i, x_i' | \boldsymbol{x}'_{i-1}) d(\hat{x}_i, x_i) dx_i \right) dx_i' \\
&= \int_{x_i'} \min_{\hat{x}_i} \int_{x_i \in \mathcal{M}} \psi_i(x_i) f_i(x_i' | x_i, \boldsymbol{x}'_{i-1}) d(\hat{x}_i, x_i) dx_i dx_i'.
\end{aligned}$$

Max-min Problem Formulation

- User i's privacy protection against the adversary's optimal inference:

$$\max \int_{x_i'} \min_{\hat{x}_i} \int_{x_i \in \mathcal{M}} \psi_i(x_i) f_i(x_i' | x_i, \boldsymbol{x}_{i-1}') d(\hat{x}_i, x_i) dx_i dx_i'$$

$$\text{s.t. } f_i^{out}(x_i' | x_i, \boldsymbol{x}_{i-1}') = 0, \forall (x_i, x_i') \in$$

$$\left\{ (x_i, x_i') | x_i \in \mathcal{X}_i^{out}, x_i' \in \mathcal{M} : d(x_i', x_i) > Q_i \right\},$$

$$\int_{x_i'} f_i(x_i' | x_i, \boldsymbol{x}_{i-1}') dx_i' = 1, \forall x_i \in \mathcal{M},$$

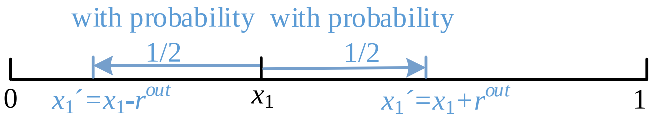
$$var : f_i(x_i' | x_i, \boldsymbol{x}_{i-1}'), \forall x_i, x_i' \in \mathcal{M}.$$

- By partitioning the location set $\mathcal{M}$ into M grids, the problem can be reformulated and solved by an LP with total complexity $\mathcal{O}(M^7)$.

Binary Approximate Obfuscation

- Instead of using continuous f_i^{in} and f_i^{out} over the location set $\mathcal{M}$, we narrow each of them to the randomization of only two location points around x_i in a normalized one-dimensional(1D) linear line $[0, 1]$.

The First User's Random Query Strategy



- The first-arriving user 1 finds the cache empty with $\mathcal{X}_1^{IN} = \emptyset$.
- User 1's two-sided random reporting strategy:

$$f_1^{out}(x_1'|x_1, \emptyset) = \begin{cases} \frac{1}{2}, & \text{if } x_1' = |x_1 - r^{out}|, \\ \frac{1}{2}, & \text{if } x_1' = 1 - |x_1 - 1 + r^{out}|, \end{cases}$$

Proposition

The optimal reporting distances and probability to maximize the first user's expected privacy is $r^{out} = \min(Q_1, \frac{1}{2})$. His maximum expected privacy gain is $\pi_1^{appr} = \min(Q_1 - Q_1^2, \frac{1}{4})$.

Latter Users' Random Query Strategy

Definition (Binary approximation for obfuscated query)

If a user i finds the cached PoI information useful (i.e., $x_i \in \mathcal{X}_i^{in}(\boldsymbol{x}'_{i-1})$), we approximate his query strategy to randomization of two symmetric points on the two sides of his real location x_i:

$$f_i^{in}(x'_i|x_i, \boldsymbol{x}'_{i-1}) = \begin{cases} \frac{1}{2}, & \text{if } x'_i = |x_i - r_i^{in}|, \\ \frac{1}{2}, & \text{if } x'_i = 1 - |x_i - 1 + r_i^{in}|. \end{cases}$$

Otherwise, if $x_i \in \mathcal{X}_i^{out}(\boldsymbol{x}'_{i-1})$,

$$f_i^{out}(x'_i|x_i, \boldsymbol{x}'_{i-1}) = \begin{cases} \frac{1}{2}, & \text{if } x'_i = |x_i - r^{out}(Q_i)|, \\ \frac{1}{2}, & \text{if } x'_i = 1 - |x_i - 1 + r^{out}(Q_i)|. \end{cases}$$

The Second User's Random Query Strategy

Proposition

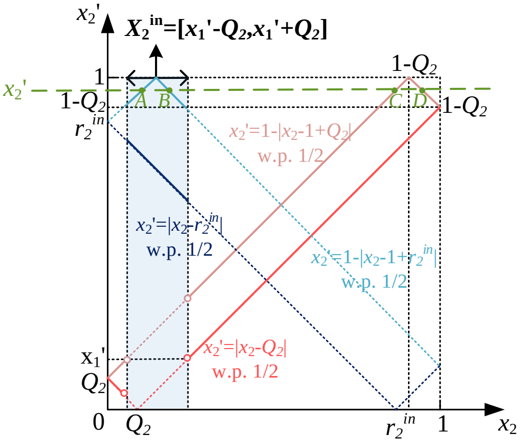
Even if user 2 finds the PoI information shared by user 1 useful (i.e., $d(x_1', x_2) \le Q_2$), it is optimal for him to misreport x_2' with distance $r_2^{in}(x_1')$ away from x_2 when querying the LBS platform, where the obfuscation distance is given by

$$r_2^{in}(x_1') = \begin{cases} 1 - Q_2, & \text{if } 0 \le x_1' \le Q_2, \\ 1 - x_1', & \text{if } Q_2 < x_1' \le \frac{1}{2}, \\ x_1', & \text{if } \frac{1}{2} < x_1' \le 1 - Q_2, \\ 1 - Q_2, & \text{if } 1 - Q_2 < x_1' \le 1. \end{cases}$$

- We manage to reduce the computational complexity of the user strategy from $\mathcal{O}(M^7)$ of the max-min problem to $\mathcal{O}(1)$.
- r_2^{in} is non-increasing in the service flexibility Q_2: the user with a greater service flexibility should choose to query the LBS with less obfuscated location.

The Second User's Random Query Strategy

- r_2^{in} is non-increasing in the service flexibility Q_2: the user with greater service flexibility should choose to query the LBS with less misreported location.



Latter Users' Random Query Strategy

Algorithm 1: Optimize r_i^{in} for any user i

Input: $\boldsymbol{x}'_{i-1}$, Q_i

Output: r_i^{in}

Initialization: $r^{out} = \min(Q_i, \frac{1}{2})$,
$\mathcal{X}_i^{in} = \cup_{j=1}^{i-1}[x'_j - Q_i, x'_j + Q_i] \cap [0,1]$, $\hat{\mathcal{X}}_i = \emptyset$,

for r_i^{in}=*0:*ϵ*:1* **do**

    for x'_i=*0:*ϵ*:1* **do**

        Compute $\hat{\mathcal{X}}_i(r_i^{in}, r^{out}, x'_i, \boldsymbol{x}'_{i-1})$ in (15)

        Compute $\hat{x}_i(x'_i) = mean(\hat{\mathcal{X}}_i)$ in (16)

        Add elements $D(\hat{\mathcal{X}}_i, \hat{x}_i(x'_i))$ to $\tilde{\mathcal{D}}$

    end

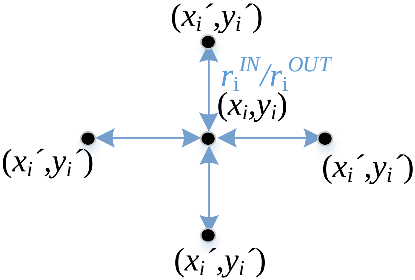
    $\pi_i^{appr}(r_i^{in}) = mean(\tilde{\mathcal{D}})$.

end

$r_i^{in} = \arg\max \pi_i^{appr}(r_i^{in})$

- Let ϵ denote the step sizes for r_i^{in} and x'_i in the iterations, respectively, then the time complexity of the algorithm is $\mathcal{O}(\frac{1}{\epsilon^2})$.

Extension to the Two-dimensional (2D) Scenario



In the 2D domain, a user's strategy is to randomize the query among four points with a fixed obfuscation distance r_i^{in}/r_i^{out} away at the four sides of his real location (x_i, y_i):

$$
\begin{aligned}
& f_i((x_i', y_i')|(x_i, y_i), (\boldsymbol{x}'_{i-1}, \boldsymbol{y}'_{i-1})) \\
=& \begin{cases}
\frac{1}{4}, \text{ if } (x_i', y_i') = (|x_i - r_i|, |y_i - r_i|), \\
\frac{1}{4}, \text{ if } (x_i', y_i') = (|x_i - r_i|, 1 - |y_i - 1 + r_i|), \\
\frac{1}{4}, \text{ if } (x_i', y_i') = (1 - |x_i - 1 + r_i|, |y_i - r_i|), \\
\frac{1}{4}, \text{ if } (x_i', y_i') = (1 - |x_i - 1 + r_i|, 1 - |y_i - 1 + r_i|).
\end{cases}
\end{aligned}
$$

Performance Evaluation

- Asymptotic optimum of the approximation scheme
- Comparison of the approximate solution with low complexity to the optimal cooperative strategy with time complexity $\mathcal{O}(M^7)$.
- Performance evaluation with single-query cooperation scheme in [Shokri et al. 2014] and the cooperation scheme with k-anonymity in [Peng et al. 2017], [L.Hu et al., 2018], [Jung et al. 2017].

Performance of the Approximation

- Asymptotic Optimum of the Approximation Scheme

Proposition

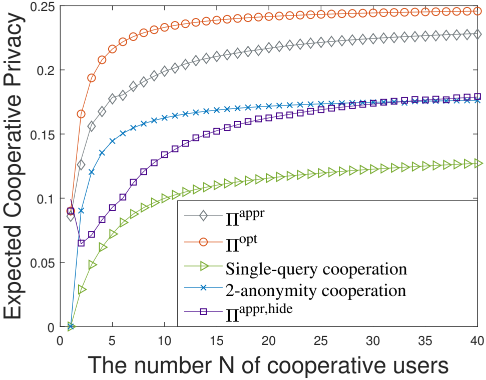
The expected privacy gain for an average user increases with the number N of cooperative users. As $N \to \infty$, the approximate obfuscated scheme is asymptotically optimal to solve the optimal privacy protection problem.

- Performance bound

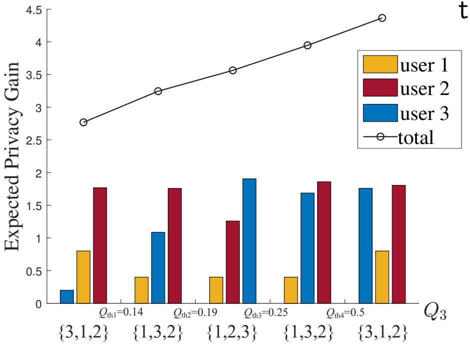
Proposition

The approximate query strategy returned by Algorithm 1 reaches at least $\frac{3}{5}$ of the maximum privacy gain in the optimal privacy protection problem.

Performance Comparison



Optimal Query Sequence among Users ($N = 3$)



When $Q_1 = 0.1$ and $Q_2 = 0.2$ are fixed, there exists four thresholds for the value of Q_3 such that:

- when $Q_3 \leq 0.14$ or $Q_3 \geq 0.5$, user 3 should query as the first one;
- when $0.14 < Q_3 \leq 0.19$ or $0.25 < Q_3 \leq 0.5$, user 3 should query as the second one;
- when $0.19 < Q_3 < 0.25$, user 3 should query as the last one.

Conclusion

- Conclusion
  - Model the privacy-preservation problem against an intelligent adversary by a two-stage game.
  - Gain privacy from service flexibility with only one or a few queries and relieve the user of reporting a great number of dummy locations to the LBS platform.
  - Propose a multi-user privacy game via cooperative obfuscation against optimal adversarial attack.
  - Propose an approximate obfuscated query scheme for cooperation with low complexity.

Thank You